

Problem

In two different ways, find the Laplace transform of

$$g(t) = \frac{d}{dt}(te^{-t}\cos t)$$

Step-by-step solution

Step 1 of 6

The function $g(t)$ is,

$$g(t) = \frac{d}{dt}(te^{-t}\cos t)$$

Method 1:

Find the Laplace transform of $g(t)$ as follows:

$$L[\cos t] = \frac{s}{s^2 + 1}$$

By frequency shifting property,

$$L[e^{-t}\cos t] = \frac{s+1}{(s+1)^2 + 1}$$

By frequency differentiation property,

$$\begin{aligned} L[te^{-t}\cos t] &= -\frac{d}{ds} \left[\frac{s+1}{(s+1)^2 + 1} \right] \\ &= - \left[\frac{(s+1)^2 + 1 - 2(s+1)(s+1)}{\{(s+1)^2 + 1\}^2} \right] \\ &= \frac{(s+1)^2 - 1}{[(s+1)^2 + 1]^2} \end{aligned}$$

Step 2 of 6

The Laplace transform of $g(t)$ is,

$$\begin{aligned} L\left[\frac{d}{dt}(te^{-t} \cos t)\right] &= s \left[\frac{(s+1)^2 - 1}{[(s+1)^2 + 1]^2} \right] - \left[\frac{(0+1)^2 - 1}{[(0+1)^2 + 1]^2} \right] \\ &= \frac{s[(s+1)^2 - 1]}{[(s+1)^2 + 1]^2} \\ &= \frac{s[s^2 + 2s + 1 - 1]}{[(s+1)^2 + 1]^2} \\ &= \frac{s[s^2 + 2s]}{[(s+1)^2 + 1]^2} \\ &= \frac{s^2[s+2]}{[(s+1)^2 + 1]^2} \end{aligned}$$

Thus, the Laplace transform of $g(t)$ is $\boxed{\frac{s^2[s+2]}{[(s+1)^2 + 1]^2}}$.

Step 3 of 6

Method 2:

The function $g(t)$ is,

$$g(t) = \frac{d}{dt}(te^{-t} \cos t)$$

$$\begin{aligned} \frac{d}{dt}[t(e^{-t} \cos t)] &= e^{-t} \cos t \frac{d}{dt}[t] + t \frac{d}{dt}[e^{-t} \cos t] \\ &= e^{-t} \cos t + t(-e^{-t} \cos t - e^{-t} \sin t) \\ &= e^{-t} \cos t - te^{-t} \cos t - te^{-t} \sin t \end{aligned}$$

$$L\left[\frac{d}{dt}[t(e^{-t} \cos t)]\right] = L[e^{-t} \cos t] - L[te^{-t} \cos t] - L[te^{-t} \sin t]$$

Step 4 of 6

Consider the Laplace following transforms.

$$L[\cos t] = \frac{s}{s^2 + 1}$$

By frequency shifting property,

$$L[e^{-t} \cos t] = \frac{s+1}{(s+1)^2 + 1}$$

By frequency differentiation property,

$$L[-te^{-t} \cos t] = - \left[\frac{(s+1)^2 - 1}{[(s+1)^2 + 1]^2} \right]$$

And,

$$L[\sin t] = \frac{1}{s^2 + 1}$$

By frequency shifting property,

$$L[e^{-t} \sin t] = \frac{1}{(s+1)^2 + 1}$$

Step 5 of 6

By frequency differentiation property,

$$\begin{aligned} L[-te^{-t} \sin t] &= - \frac{d}{ds} \left[\frac{1}{(s+1)^2 + 1} \right] \\ &= \left[\frac{((s+1)^2 + 1)(0) - 2(s+1)}{\{(s+1)^2 + 1\}^2} \right] \\ &= \frac{-2(s+1)}{[(s+1)^2 + 1]^2} \end{aligned}$$

Step 6 of 6

Hence, the Laplace transform of $g(t)$ is,

$$\begin{aligned}
 L\left[\frac{d}{dt}(te^{-t}\cos t)\right] &= \frac{s+1}{(s+1)^2+1} - \frac{[(s+1)^2-1]}{[(s+1)^2+1]^2} - \frac{2(s+1)}{[(s+1)^2+1]^2} \\
 &= \frac{s+1}{(s+1)^2+1} - \left[\frac{s^2+1+2s-1+2s+2}{[(s+1)^2+1]^2} \right] \\
 &= \frac{(s+1)(s^2+2+2s) - (s^2+4s+2)}{[(s+1)^2+1]^2} \\
 &= \frac{s^3+2s+2s^2+s^2+2+2s-s^2-4s-2}{[(s+1)^2+1]^2} \\
 &= \frac{s^3+2s^2}{[(s+1)^2+1]^2} \\
 &= \frac{s^2[s+2]}{[(s+1)^2+1]^2}
 \end{aligned}$$

Thus, the Laplace transform of $g(t)$ is $\boxed{\frac{s^2[s+2]}{[(s+1)^2+1]^2}}$.